

Depression Prevention by Mutual Empathy Training: Using Virtual Reality as a Tool

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ABSTRACT

The increasing number of diagnosed anxiety and depression has been aggravated by the COVID-19 pandemic, in addition to social alienation in modern life. Therapeutic intervention for depression using serious games is gaining popularity in recent years. However, the prevention for depression among people with sub-health status has not gained enough attention, and the avoidance of relapse depression under post-pandemic social alienation is ever more challenging. This paper proposes a novel approach to prevent depression and its recurrence through empathy training using interactive virtual reality game applications. This study explores the possibility that VR games, as an empathy machine, can grow into a social network with interpersonal mutual and empathetic support that benefits both prevention and intervention methods for depression.

Index Terms: Human-centered computing—Human-computer interaction (HCI)—Empirical studies in HCI; Human-centered computing—Human-computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Due to the pandemic of COVID-19, more people are at risk of anxiety or depression and may respond more strongly to the stress [32, 37]. Fear and anxiety about the pandemic and the new social alienation lifestyle could cause overwhelmingly strong emotions and even depression or relapse of depression [2, 33]. Studies [26] have suggested that overly active and automatic empathetic reaction is one of many causes of depression; meanwhile, the lack of empathy from social environment because of the stigma to the patients is another factor [36]. Therefore, a portable technology that is affordable and easy to access is highly needed to reduce depression by balancing empathy: promoting external empathy and training internal empathy.

Game-based solutions can be beneficial in improving depression and bonding families and release stress by having entertainment activities together [7, 18, 21]. Advanced gaming tools, especially virtual reality (VR), have shown to be able to help improve or facilitate early intervention for mental disorders [24, 28, 29]. Meanwhile, VR can offer stronger emotional experience with a better concentration than traditional visual/audio media because of the ability to provide an immersive presence in a safely controllable virtual environment. Researchers have found the advantages of VR to be an empathy machine [12].

By facilitating the communication and connection between patients and others, the proposed study seeks a better social environment for depression, which will eventually help prevent depression onset and relapse.

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2 BACKGROUND

As a long-term ongoing research project, we first conducted a systematic review [19] on using head-mounted VR (including interactive application and 360-degree video simulation) for promoting empathy or increasing public awareness, not targeting any particular symptoms or social issues. 427 articles were sorted through and 10 were identified relevant. This section includes briefs of findings of the literature that is related to the current study.

2.1 Empathy Training

Psychological studies found that social support is one of the key factors in helping patients recover from depression, in addition to counseling sessions and medications [3, 11, 20]. To cultivate social support in the social network and promote public awareness, the understanding of the basic symptoms that lead to empathy should be raised [30]. On the other hand, the overly amount of empathetic reaction caused by the cognitive distortions of the patients is considered one of many causes of depression [36]. Therefore, the need for appropriate empathy on both sides calls for carefully designed studies to address empathy training for depression.

2.2 VR as an Empathy Machine

As more and more researchers become aware of VR as “an ultimate empathy machine” [14, 25], adopting VR for learning empathy is widely recognized in medical education [1, 23, 34]. The immersion and embodiment of VR applications have been proven to help medical students visualize and understand patients’ feelings to help them better. Pilot studies in medical education have reported cultivating empathy for older people [6] and nausea and vomiting management [35], among others. Only two games using VR designed for increasing self-compassion and empathy for depression [5, 9] were identified by the review. The study [9] made use of recorded VR 360-degree videos, and study [5] used real-time camera and a virtual-self to simulate real-life therapy sessions. They both showed positive results in raising empathy levels. However, since both of them are not using immersive interactive game scenes, they may not have fully revealed the best potential and advantages of using VR applications. That is to say, to the best of our knowledge, no study had designed true immersive interactive VR applications to promote empathy in depression, nor had they investigated the effectiveness and impact of the applications.

2.3 Our Contribution

In this project, we propose a study aims to raise mutual understanding and trains appropriate empathy level. The approach not only assists depression intervention for patients in an indirect way, but also leads to the prevention of possible depression onset in people with risk by letting them know the symptoms and also the relapse of recovered patients by promoting more social supports, as well as training their own empathy level.

3 METHODS

As very few studies have done similar work before, we designed a study with an interactive VR prototype in the hope of proposing a framework for developing future VR applications for similar



Figure 1: An example scene of interaction with NPC

purposes. We plan to test the hypothesis based on validated measurements in empathy training [4, 8, 17], as well as embedded in-game performance measurements.

3.1 Objectives

A VR empathy game application named “Schwer” was developed. The targeted audience includes a). the patients diagnosed with depression, b). caregivers such as family and friends.

The proposed study aims to establish cognitive empathy and mutual understanding of depressive disorders in patients and caregivers. By address the pressure of social interaction and relieve the empathic distress that the caregivers are likely to face, the study uses the VR application to train appropriate empathy level. This could help people at risk of getting depression or relapse after recovery to take a step back and re-evaluate their mental condition, which will ultimately benefit depression prevention.

3.2 VR Application Design

The prototype has been developed by a group of undergraduate students with advice from two clinical counselors and two patients with major depression along the course of development and implementation. Feedback and comments are provided in regular remote meetings and demo presentations. All the objectives proposed can be quantitatively measured.

3.2.1 System Architecture

The game application is VR-based, fully immersive, with narrative and interaction designs instead of a recorded VR 360 video or a simulation of an intervention session where a virtual therapist is presented. This is to fully utilize VR devices’ presence and immersion experience and motion feedback for promoting empathy. Unity game engine is used to develop the game, and the targeted platforms include two major popular VR headsets: Oculus and HTC Vive, and theoretically any headsets that are compatible with SteamVR. Patients with depression were consulted to provide insights on artworks, environments, storylines, and level maps/events, to provide an opportunity for them to express their feelings metaphorically. Background music and sound effect serve as real-time feedback to the accomplishment of quests and performance measurements.

3.2.2 Virtual Environment

Game mechanism. The VR application is a first-person view adventure game. The degree of distress from patients with depression is represented by the dynamically changing stamina levels that decide how far the player can teleport for each step, the darkness and brightness of the scene, and the growing or shrinking restricted area covered by water. The progress of the game is indicated by the water level of the game world and the lighting-up area of a mini brain map.



Figure 2: Examples of the UIs.

The player interaction with the non-player characters (NPCs) and attempts to accomplish the quests will push forward the progress and narrative. The player will have to go through a maze puzzle to collect different supplies to finish a task, including medication and physical supplies, to progress through the journey.

Characters. Three types of NPCs offer different help or feedback when appropriately interacted. They map to different social roles in the real world with varying power and limitations in the virtual world: a). The Wise: Corresponding to the roles of “psychologist” and “professional counselor” in the real world; b). The Sage: Corresponding to the roles of “family” and “friend” in real world as social support; c). The Fool: Corresponding to the role of people who “do not understand depression” or “depression is just thinking too much” in the real world. They will provide professional guidance, provide stamina support, or barricade the progress, respectively. The Fig. 1 shows an example scene of interaction with the Wise NPC.

Self-perspective measures. The Form of Self-Criticizing /Attacking & Self Reassuring Scale (FSCRS) [4] evaluates the feeling of worthless, useless, or inferior. The improvement of this measure reflects the cognitive behavior that we aim to reshape in the third objective. A standard FSCRS scale is embedded into the dialogues between the NPCs and players (an example is shown in Fig. 1).

The main character player has different attributes that will dynamically change based on the player’s choices. The progress of these choices will lead to different interactions and feedback. These attributes include stamina, intelligence, interaction ability, and emotional capacity. The left screenshot of Fig. 2 shows a mini brain map and its variation set, where lightened areas and water level indicate intelligence attribute and emotional capacity. The right screenshot of Fig. 2 shows the stamina bar that will affect the teleporting distance, staying as close to middle as possible will translate into the farthest teleport range.

The game starts with the desolation stage, where the world is covered by water of desperate. Everything is destructed and deserted. The scene is dark, and the teleporting distance of the player is limited depending on the stamina and intelligence of the player. The player needs to explore the game world and uncover unknown routes in puzzles, and perform scavenger hunting according to the clues given by the NPCs. The goal for the player is to survive through puzzle challenges, collect correct supplies guided by the Sage NPC and the Wise NPC, and avoid the Fool NPC’s misleading or insulting words. The water level will shrink as the player advances, and the brain map will be lightened up one area by another to indicate the progress. After stamina and intelligence attributes are successfully restored, the player will enter the transition scene.

In the self-healing transition scene, the player detaches from the body and spectate him/herself watching a campfire starting and putting out. A fire fox will be born from the campfire representing “the hope”.

Finally, in the civilization stage, the player starts reconstruction. A peaceful botanical garden will be built; the view becomes bright, the weather and music turn to be harmonious.

Fig. 3 shows screenshots of different stages. The upper-left screenshot is the map where city building blocks being covered or separated



Figure 3: City buried in sea and the dark environment (upper-left), transition scene (upper-right), brain map at beginning (bottom-left), brain map after quests (bottom-right)

by water under dark and foggy weather, depicting the feelings of sadness, frustration and more as the symptoms of depression. The upper-right screenshot is the transition scene where the player detaches from the body and accompanies him/herself to watch the campfire. The bottom two screenshots show the differences in brain map and environment tone between the desolation and civilization stages.

3.2.3 Psychology-laden training

Colors, rewards, and positive reinforcement are used to help build connections between patients and caregivers. Depression often causes cognitive impairment and emotional impairment, which significantly impacts the patient's functioning and quality of life [31]. Furthermore, patients often lack the willingness and ability to ask for help even when necessary. Inspired by the concept used in cognitive behavioral therapy (CBT) [15, 16, 27] and mindfulness-based intervention [10, 13, 22], and also under the advice of counselors, a game flow is designed for training cognitive behavior with an upward spiral process, which is depicted by Fig. 4.

1. In the beginning, the subject lacks the willingness and ability to ask for help;
2. The subject learns how to express negative feelings and ask for help from an NPC to survive in the game world;
3. The subject gains the feeling of control and self-efficacy from the game feedback, e.g., getting instruction from one of the NPCs;
4. The subject learns that asking for help is an option in reality that she should not be ashamed of, and that the behavior benefits both sides;
5. Help and feedback from caregivers in real-world will reduce the subject's feeling of powerlessness and self-criticism;
6. The cognitive behavior has been reshaped, and the subject gets into another round of enhancement or reshape.

The first three items match the puzzle and challenges presented at the desolation stage, while the last three items match the reconstruction process at the civilization stage.

4 DISCUSSION

The VR application prototype has been developed as expected to demonstrate the concept. Throughout the design and development stage, two counselors have provided feedback to level design ideas. Game design students and two patients with depression have



Figure 4: The upward spiral of cognitive behavior

playtested, and feedback and suggestions have been incorporated into the application.

As mentioned in section 3.2.2, the FSCRS is embedded in the gameplay to measure and dynamically change game progress. Meanwhile, a pre- and post- questionnaire is designed to use the Interpersonal Reactivity Index (IRI) [8] to evaluate the empathy level change of the participants.

The face-to-face experiments were unable to finish as planned due to the pandemic of COVID-19, but we managed to recruit a few participants to playtest the application and collected feedback. All participants expressed gaining understanding of the mental world of depression and showed interest in participating in future experiments.

5 CONCLUSION AND FUTURE WORK

In the current study, we proposed a VR application to train mutual empathy and build social support for depression, people at risk of the disorder or relapse. We hope to contribute to the field because not only should the social environment pay attention and learn to provide support to the ones who suffer, but the patients should also be aware of the help they may get, meanwhile cognitively change their behaviors and give others feedback to build mutual and healthy connection. VR, as an empathy machine, will be beneficial to both prevention and intervention methods for depression.

Ultimately, we would like to publish the game to multiple platform app stores once the study is done. A full game version will be tailored for public usage to foster a social support community for depression. A link to the game trailer can be found below.

<https://youtu.be/bpEupc3QVps>

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